Host-Conditioned Spatial Structure of Canopy Mortality in Finland's Boreal Forests

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Abstract

Finland's boreal forests are experiencing a documented increase in non-harvest tree mortality [1], and the spatial arrangement of dead trees encodes the processes that generate it. We analyse 698,223 individual dead-tree locations (detected by a U-Net segmentation model from 2023 aerial imagery across a national-scale sample of seven boreal super-sites, ∼60.4–67.8°N) as a *host-conditioned point process*, asking whether mortality clusters more than forest host availability alone explains. Conditioning on a pre-mortality (2021) multi-source forest-inventory host surface, we find that mortality intensity increases with both spruce and total growing-stock volume (strong host-tracking), and that, beyond this host structure, mortality retains a *modest but genuine residual aggregation* that holds from the sub-kilometre scale out to at least 3 km: the inhomogeneous L-function lies above a host-conditioned simulation envelope in all seven blocks even under a richer null (spruce + total volume + stand edge), and the same exceedance appears whether the process is taken at the cluster-centroid or the individual-tree level: a local-contagion signature on top of host availability that is not an artefact of the clustering unit. We establish these claims with a reusable falsification-and-validation harness for morphometrics derived from black-box detection models: a covariance-preserving null with mixture-model discreteness diagnostics, clustering scale and shape-filter stability gates, detector-confound propagation (cluster "type" predictable from detector confidence alone at AUC = 0.727), and purposive cluster-sample inference (effective sample size = 7 blocks, not 14,582 clusters), which also shows that a discrete shape-"typology" of the clusters is an artefact of clustering scale, filter thresholds and detector behaviour rather than ecology. We therefore report spatial *structure*, not agent-labelled types: lacking species, dating, and ground-truth layers, disturbance-agent attribution is withheld. A marked-process test against an external disturbance-agent layer is the natural next step.

Keywords: boreal forest mortality; spatial clustering; DBSCAN; Clark–Evans index; morphological gradient; cluster shape metrics; clustering-scale sensitivity; Finland

1 Introduction

Boreal forests store roughly 30% of terrestrial carbon and support exceptional biodiversity, yet they are undergoing accelerating disturbance as climate change intensifies drought, windstorms, and pest outbreaks [2; 3]. In Finland, non-harvest tree mortality has increased substantially in recent years [1], a trajectory that demands systematic monitoring. Understanding which ecological processes drive this mortality, and where they concentrate across the landscape, is a prerequisite for adaptive management and carbon-flux modelling.

The spatial arrangement of dead trees encodes information about the process that killed them. Bark-beetle outbreaks typically generate dense, spatially contagious patches as infestation spreads from tree to tree [4; 5]. Wind damage, by contrast, tends to produce elongated, directionally oriented swaths aligned with storm tracks [6; 7]. Drought-induced mortality can produce more scattered, diffuse patterns tied to soil-water gradients [8]. These morphological *fingerprints* are in principle recoverable from the spatial arrangement of dead-tree locations even when individual tree species and cause-of-death records are unavailable.

High-resolution aerial imagery now enables the detection and georeferencing of individual dead trees over large areas. The 2023 aerial sample used here produced 698,223 dead-tree locations in ETRS89/TM35FIN (EPSG:3067) coordinates, distributed across Finland from the southern boreal zone to Lapland, a dataset large enough to support robust statistical characterisation of cluster morphology at scales ranging from the individual stand to the national sample.

This paper makes three contributions, one empirical and two methodological:

1. **Host-conditioned spatial characterisation of boreal canopy mortality.** Treating 698,223 individual dead-tree detections as a point process (their mapped locations analysed as a spatial pattern) and conditioning on a pre-mortality forest-inventory host surface, we show that mortality intensity tracks host availability (it rises with both spruce and total growing-stock volume) and yet retains a modest, genuine *residual aggregation* that survives a richer host-conditioned null (spruce + total volume + stand edge) from the sub-kilometre scale out to at least 3 km, in all seven blocks and at both the cluster-centroid and individual-tree level: a local-contagion signature on top of host-tracking, established with inhomogeneous second-order statistics and host-conditioned simulation envelopes.

2. **A falsification-and-validation harness for morphometrics from black-box detectors.** A reusable, model-agnostic protocol that asks whether spatial statistics computed on segmentation outputs reflect ecology or the instrument: a detector-confound test using the model's own confidence (cluster "type" predictable at AUC = 0.727), detection-error propagation through the full pipeline, clustering-scale and shape-filter stability gates, a covariance-preserving null with mixture-model discreteness diagnostics, purposive cluster-sample inference (effective sample size = the 7 sampling blocks), and an external check against forest-inventory species composition. Applied here it overturns most candidate signals and certifies the host-conditioned residual.

3. **A demonstration that cluster shape alone does not attribute disturbance agent.** A candidate two-type morphological "typology" of the clusters dissolves under the harness: it is not invariant to the clustering radius, its prevalence is an artefact of the aspect-ratio filter, and (aspect-ratio-matched) it carries no directional point-process signal beyond hull shape; an external check against forest-inventory species composition confirms the morphological gradient does not even track the stand species mix. We therefore characterise spatial structure rather than infer agents from patch geometry, a transferable caution for the aerial mortality-mapping literature.

Throughout, we deliberately withhold disturbance-agent labels. Morphological associations with wind, bark beetle, or other agents are testable hypotheses, not findings: confirmed attribution would require species records, mortality dates linked to known events (the available `alku_aika` field records flight-strip acquisition time, not mortality date), and multi-temporal imagery, none of which are present in the current dataset. We return to specific attribution caveats, including the absence of any directional signal in the elongated type, in the Discussion.

2 Related Work

European forest disturbance ecology and mapping. Europe's forests have been subject to intensifying disturbance regimes over recent decades, driven by the interaction of climate change with biotic and abiotic agents [9; 10]. Senf and Seidl [11] produced the first continent-wide map of forest disturbance regimes using satellite time series, revealing strong latitudinal gradients in the relative importance of wind, bark beetles, and fire. Allen et al. [3] documented globally the accelerating role of drought and heat as pre-disposing stressors that increase forest vulnerability to secondary agents. These landscape-scale perspectives highlight the need for methods that can characterise disturbance not just by area affected, but by the spatial structure of affected patches.
Bark beetle ecology and spatial signatures. The Eurasian spruce bark beetle (*Ips typographus*) is the dominant biotic disturbance agent in boreal and subalpine Europe [5]. Raffa et al. [4] showed that bark-beetle eruptions are governed by cross-scale feedbacks: fine-scale tree-to-tree contagion at the stand level, mediated by pheromone plumes and resource depletion, scales up to landscape-level epidemics when predisposing conditions (drought, windthrow) are met. Jönsson et al. [12] analysed epidemic dynamics of *Ips typographus* in Sweden, identifying that outbreaks initiated in windthrown timber and then spread as self-reinforcing patch growth. Kautz et al. [13] demonstrated that high-resolution empirical host-tree distributions can be used to simulate outbreak spread, predicting the compact, dense patch morphology that characterises beetle kills. Wichmann and Ravn [14] documented how storm-felled timber from the 1999 Danish windstorm acted as a beetle breeding reservoir, amplifying subsequent outbreaks and leaving spatially contagious kill patches.
Wind damage in Fennoscandia. Wind is the leading abiotic disturbance agent in northern European forests [15]. Gardiner et al. [6] reviewed how silvicultural history, stand structure, and edge exposure interact to determine windstorm susceptibility, noting that clearcut edges and monoculture Norway spruce stands are partic-

ularly vulnerable. Gregow et al. [7] specifically modelled combined wind and snow-loading risks in Finland under current and projected future climate, finding that snow-induced stem damage concentrated in central and eastern Finland may increase with warming winters. The elongated, directional morphology of wind-felled areas has been noted in several Fennoscandian studies as a diagnostic feature distinguishing wind damage from beetle-induced mortality.

Spatial statistics methods. The Clark–Evans statistic [16] remains the canonical index of point dispersion within bounded study regions. Donnelly [17] derived an analytical edge correction that removes the downward bias introduced by boundary effects in finite study areas, making the corrected statistic suitable for small clusters with irregular convex-hull boundaries. Anselin [18] introduced Local Indicators of Spatial Association (LISA), enabling the identification of statistically significant spatial clusters and outliers at multiple scales. Wiegand and Moloney [19] provide a comprehensive treatment of spatial point-pattern analysis methods for ecological applications, including the mark correlation function, Ripley's K, and related second-order statistics that complement the first-order metrics used here.

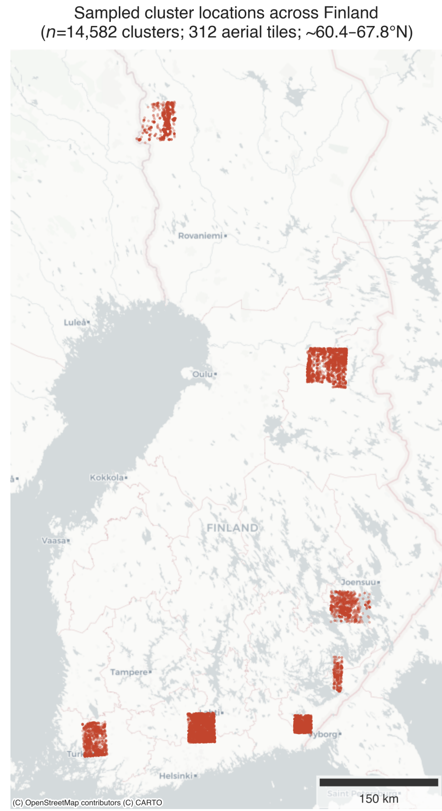

Figure 1: Sampling footprint: all 14,582 cluster centroids (from 312 aerial tiles) on a national basemap. The tiles form discrete study blocks spanning ∼60.4–67.8°N (southern boreal zone to Lapland) and the full longitudinal width, rather than contiguous national coverage.

3 Methods

3.1 Dataset

The dataset comprises 698,223 dead-tree locations produced internally by applying the individual-tree mortality U-Net segmentation model of Junttila et al. [1] to high-resolution aerial imagery acquired in 2023; the underlying detection dataset is not yet publicly released. The detections come from 312 aerial image tiles distributed across Finland, spanning roughly 60.4–67.8°N and 21.6–29.6°E, from the southern boreal zone to Lapland, in discrete latitudinal bands rather than continuous wall-to-wall coverage. Coordinates are provided in ETRS89/TM35FIN (EPSG:3067), a conformal transverse Mercator projection with metric units, appropriate for distance-based spatial analyses. The dataset is therefore a *national-scale but spatially discontinuous sample*: reported cluster prevalences and densities are sample statistics across the surveyed tiles and should not be read as an exhaustive national census. Figure 1 shows the sampling footprint, discrete study blocks from the southern coast to Lapland.

Sampling design. Formally the data constitute a *purposive two-stage cluster sample*, not a probability sample of Finnish forest and not a national census. The primary units are 7 spatially contiguous blocks ("supersites"), each a mosaic of adjacent 6 km tiles spanning 25–54 km and separated from its neighbours by 84–348 km (median 149 km); the secondary units are the 6 km map-sheet tiles; and within each tile all detectable dead-tree canopies are enumerated (a within-tile census). The blocks were compiled operationally rather than drawn under a known probability mechanism, so design-based inference to a national population is not available and we adopt a model-based, block-conditional stance throughout. Crucially, the 14,582 clusters are not independent draws: they are nested within 7 blocks that share imagery campaign, climate band, host composition, and detector behaviour, so the effective sample size for any between-block or national statement is governed by the number of blocks (7), not the number of clusters. We therefore restrict inferential claims to within-cluster and within-block morphology, report prevalences as conditional on the sampled blocks (and on the eps and AR choices; Section 4.7), and treat the phrase "national-scale" as describing the latitudinal *extent* of the sample, never national coverage or national population estimates.

3.2 Clustering

Density-based spatial clustering was performed using DBSCAN [20] with neighbourhood radius eps = 20 m and minimum neighbourhood size min_samples = 3. The eps = 20 m threshold was chosen to capture within-stand tree-to-tree spread typical of both bark-beetle and wind disturbance, while separating clusters arising from distinct events. The 43,539 raw clusters were post-filtered to retain only those satisfying:

- $n \geq 5$ trees (excludes trivially small groups),
- area $\geq 100\,\mathrm{m}^2$ (minimum ecologically meaningful patch size),
- aspect ratio ≤ 10 (removes linear artefacts such as road corridors).

This yielded 14,582 clusters for analysis.

3.3 Spatial Metrics

Each cluster was characterised by the following spatial metrics.

Within-cluster nearest-neighbour distance (NND). For a cluster containing n points, the NND for point i is

$$d_i = \min_{j \neq i} \|\mathbf{p}_i - \mathbf{p}_j\|, \tag{1}$$

and the cluster-level summary statistics are $\overline{\mathrm{NND}} = n^{-1} \sum_i d_i$ and $\sigma_{\mathrm{NND}} = \sqrt{n^{-1} \sum_i (d_i - \overline{\mathrm{NND}})^2}$.

Clark–Evans index with Donnelly edge correction. The Clark–Evans index compares the observed mean NND to its expectation under complete spatial randomness [16]:

$$R = \frac{\overline{\mathrm{NND}}}{E[\bar{r}]}, \tag{2}$$

where $R < 1$ indicates clustering and $R > 1$ indicates dispersion. For small, bounded clusters the uncorrected expectation $E[\bar{r}] = 0.5\sqrt{A/n}$ is negatively biased by boundary effects. Following Donnelly [17], we use the corrected expectation

$$E[\bar{r}] = 0.5\sqrt{\frac{A}{n}} + \left(0.0514 + \frac{0.041}{\sqrt{n}}\right)\frac{P}{n}, \tag{3}$$

where A is the convex-hull area of the cluster and P is its perimeter.

Fractal dimension. Boundary complexity is quantified by the box-counting fractal dimension:

$$D = -\lim_{\epsilon \to 0} \frac{\log N(\epsilon)}{\log \epsilon}, \tag{4}$$

where $N(\epsilon)$ is the number of boxes of side length ϵ that intersect the cluster's convex hull. In practice D is estimated as the slope of $\log N(\epsilon)$ versus $\log(1/\epsilon)$ over a range of scales expressed as fractions of the hull's bounding-box diagonal. Because the metric is computed on the convex hull rather than the raw point set, it measures the area-filling complexity of the hull boundary; for the small clusters that dominate the dataset (median eight points) it is correspondingly coarse, and we treat it as a descriptive shape index rather than a true fractal estimate.

Compactness (form factor).

$$C = \frac{4\pi A}{P^2}, \tag{5}$$

where A and P are the convex-hull area and perimeter. $C = 1$ for a circle and decreases as the shape becomes more elongated or irregular.

Aspect ratio. The aspect ratio is the ratio of the major to minor axis of the minimum-area bounding rectangle:

$$\mathrm{AR} = \frac{\ell_{\max}}{\ell_{\min}}. \tag{6}$$

Alpha compactness.

$$\mathrm{AC} = \frac{\mathrm{area}(\mathcal{A}_\alpha)}{\mathrm{area}(\mathcal{H})}, \tag{7}$$

where $\mathcal{A}_\alpha$ is the alpha shape of the point set and $\mathcal{H}$ is its convex hull. Values near 1 indicate dense, convex clusters; lower values indicate concave or elongated clusters with interior voids.

Core-to-edge ratio. The ratio of the convex hull's interior area to its total area, where the interior is the hull eroded inward by a fixed 1 m buffer: $\mathrm{area}(\mathcal{H} \ominus 1\,\mathrm{m})/\mathrm{area}(\mathcal{H})$. Because the erosion width is fixed while clusters vary in size, this ratio is partly a proxy for absolute cluster size (small clusters lose proportionally more area), which should be borne in mind when interpreting it.

Orientation. PCA is applied to the xy coordinates of cluster points; the orientation angle is

$$\theta = \arctan\left(\frac{v_y}{v_x}\right), \tag{8}$$

where $\mathbf{v} = (v_x, v_y)$ is the first principal eigenvector. θ is folded into $[0°, 180°)$ for use as axial data in orientation analysis.

3.4 Morphological Gradient and Candidate Partition

K-means clustering was applied to the standardised feature vector (fractal_dim, clark_evans, aspect_ratio) using $k \in \{2, \ldots, 7\}$. The optimal k was selected by jointly evaluating the silhouette coefficient, Calinski–Harabasz (CH) score, and Davies–Bouldin (DB) index. At $k = 2$ these were silhouette = 0.401, CH = 8767, DB = 1.073, the best combined performance across the tested range.

To assess whether the observed silhouette score reflects genuine structure rather than the covariance of correlated marginals, we compared it against a *covariance-preserving* null: 999 surrogate datasets were drawn from a single multivariate Gaussian whose mean and full covariance matrix match the standardised feature matrix, and $k = 2$ K-means was refitted to each. This is a stricter null than independent column-shuffling (which destroys all inter-feature correlation and is therefore beaten by any correlated data); it asks specifically whether the data are *more clustered than a single elongated Gaussian cloud*. The observed silhouette (0.401) exceeded the null mean (0.321) and 95th percentile (0.324), giving $p = 0.001$.

Because a significant silhouette does not by itself establish that two discrete classes exist, we additionally tested for discreteness. Gaussian-mixture model selection by BIC over $k \in \{1, \ldots, 7\}$ and the modality of the first principal component (which captures 60% of feature variance) jointly indicate that the morphological features form a continuous gradient rather than two separated modes (Section 4.2). We therefore treat the $k = 2$ partition as an interpretive summary of the dominant compactness–elongation axis, not as evidence of a natural class boundary.

3.5 Orientation Analysis

Cluster orientations $\theta \in [0°, 180°)$ are axial data and must be doubled before applying circular statistics [19]. The Rayleigh test for uniformity was applied to the doubled angles $2\theta \in [0°, 360°)$ using pycircstat. The mean resultant length $r = \|\sum \exp(i \cdot 2\theta_j)\|/n$ and its associated p-value quantify whether orientations depart from a uniform (no preferred direction) distribution. Results were stratified by type.

3.6 Spatial Autocorrelation

Global Moran's I was computed for four cluster-level metrics (fractal_dim, clark_evans, aspect_ratio, compactness) at five spatial thresholds (200, 500, 1000, 2000, 5000 m) using a random subsample of $n = 2{,}000$ clusters stratified by block, which preserves the regional structure while keeping the dense distance-band weight matrix tractable. A distance-band contiguity at threshold δ ($d_{ij} \leq \delta$) was used, with the weights row-standardised ($\sum_j w_{ij} = 1$). With 20 simultaneous tests we applied a Bonferroni-corrected significance threshold $\alpha^* = 0.05/20 = 0.0025$; significance is assessed with the analytical (normal-approximation) p-value, since a 999-permutation p_{sim} floor of 10^{-3} is the coarsest resolution that can reach α^* and a 199-permutation test (floor 5×10^{-3}) cannot. To test whether any autocorrelation is an artefact of upstream detection error rather than ecology, we additionally propagated a 15% random false-negative detection perturbation through the metric computation over 100 realizations and recorded the fraction preserving the sign and Bonferroni significance of I at each metric $\times$ threshold (a detection-robustness gate). Because the sample comprises discrete latitudinal bands separated by large gaps, the distance-band weights connect only within-band neighbours at all thresholds tested; the resulting statistic is therefore an estimate of *within-band* autocorrelation, and we report per-threshold neighbour counts to expose how thin the short-distance support is.

Local Moran's I (LISA; Anselin 18) was computed to identify statistically significant spatial clusters (high–high and low–low) and spatial outliers in fractal dimension.

3.7 Landscape Connectivity

A proximity graph was constructed by connecting all cluster centroids within 200 m [21]. The reach of a cluster is defined as the number of other clusters reachable within the connected component to which it belongs. Betweenness centrality was computed to identify bridge clusters whose removal would fragment the network. Reach distributions were compared between types using the Mann–Whitney U test.

3.8 Held-Out Classifier

To test whether the $k = 2$ partition is recoverable in a feature space not used to define it, we trained a Random Forest classifier using only features *not* input to K-means: compactness, alpha_compactness, core_to_edge_ratio, nnd_mean, nnd_std, orientation, and reach_200m. (We note that `orientation` is axial data on $[0°, 180°)$ entered here as a linear feature, so the tree splits cannot exploit its circular structure; given its low importance below, this does not affect the conclusions.) Spatial block cross-validation (GroupKFold, 5 folds, 100×100 km grid, 12 blocks, mean 1215 clusters/block) was used to prevent spatial leakage. We caution that the blocks are size-imbalanced, the five largest hold roughly four-fifths of the clusters, so the tight fold-to-fold AUC range understates true spatial variability. SHAP values [22] were computed to assess feature importance.

3.9 Stability Gates

These gates are deliberate attempts to *falsify* our own two-type result: a genuine ecological partition should survive them, whereas one manufactured by an analysis choice should break. To test whether the $k = 2$ partition is a genuine structure or an artefact of the pipeline's free parameters, we re-ran the entire pipeline from the raw points under four gates.

(G1) Scale stability. We re-clustered the raw points at eps $\in \{10, 15, 20, 25, 30, 40\}$ m and, at each scale, recomputed all shape metrics, re-applied the filters, and re-fitted $k = 2$ K-means. A genuine class should be eps-invariant in prevalence and in its feature-space centroid; an artefact of detection chaining rescales with eps.

(G2) Aspect-ratio filter, from raw. The AR ≤ 10 filter is applied to the very axis on which K-means clusters, so it cannot be tested on the already-filtered table. We instead recomputed metrics on all clusters at eps = 20 m with *no* AR cap, then applied caps $\in \{5, 7, 10, 15, 20, \infty\}$, re-fitting K-means at each and tracking type prevalence, the within-Type 0 AR distribution, and the adjusted Rand index (ARI) of membership against the AR $\leq$ 10 partition.

(G3) Directional second-order structure. Hull aspect ratio is a first-order summary and cannot distinguish a linear (wind-swath) point arrangement from an elongated but space-filling patch. For each cluster we computed the point-scatter linearity $\ell = \lambda_1/(\lambda_1 + \lambda_2)$ from the PCA eigenvalues and a pairwise-displacement axial resultant, and compared Type 0 against Type 1 *within matched hull-AR bins.*

(G4) Detection confound. The upstream detector provides a per-detection confidence (`max_value`) not otherwise used. We tested whether per-cluster detection covariates (mean/SD confidence, n, density) predict type (5-fold logistic AUC), and whether modest false-negative perturbation (dropping 10–20% of detections per cluster and re-fitting) changes type assignments.

(G5) External species validation. To break the circularity of reading an agent off the shape we clustered on, we attached an external, non-shape variable to each cluster: the pre-mortality (2021) MS-NFI growing-stock volume of spruce, pine and birch, sampled at the cluster centroid (EPSG:3067, nodata-masked; clusters off forest host dropped). We then asked whether the morphological gradient tracks the species mix of the stand that died, with the block as the replicate: a within-block Spearman correlation between aspect ratio and each species fraction combined across the seven blocks by a sign test, and a per-block comparison of species fractions between the two morphological endpoints. No per-cluster p-value is reported (it would be pseudo-replicated across ∼2,000 clusters per block).

3.10 Host-Conditioned Point-Process Analysis

To ask whether mortality clusters *more than forest host availability alone predicts*, we replace the homogeneous-CSR null, meaningless over heterogeneous forest, with an *inhomogeneous* null whose first-order intensity is driven by host. In plain terms, the baseline we compare against is not "dead trees scattered evenly across the landscape" but "dead trees in proportion to how much host forest is present"; any clustering beyond that baseline (what we call *residual aggregation*) is the part not explained by where the forest is. The host surface is the Multi-Source National Forest Inventory (MS-NFI) spruce growing-stock volume from *2021*, i.e. *before* the 2023 mortality: a 2023 host layer would be endogenous, since dead spruce drops out of the standing-volume estimate. The observation window is the union of the 6 km map-sheet tiles restricted to forest land (MS-NFI land-class mask), not the tile bounding boxes (which include lakes) and not "spruce > 0" (which would conflate covariate and window). We model the intensity $\lambda(u)$ by a pixel-Poisson log-linear fit on the forest grid (covariates: spruce volume, total growing-stock volume, distance to stand edge) and assess second-order structure with the inhomogeneous K- and L-functions and pair-correlation, pooled across tiles with translation edge correction. Significance is by *host-conditioned* simulation envelopes (99 realisations from the fitted inhomogeneous null), not the CSR contrast. The estimator was validated on simulated inhomogeneous-Poisson patterns ($L_{\text{inhom}} - r \approx 0$). We run two independent implementations: a tile-pooled inhomogeneous pair-correlation/L (intensity normalised per tile; the ring estimator on 6 km tiles is reliable to $r \lesssim 1$ km, $\approx$ window/4), and a per-block L_{inhom} in `spatstat` with the intensity fitted globally (pooled Poisson across forest cells), cumulative translation-corrected and therefore variance-stabilised, on the 25–54 km block windows, which extends the reliable range to $r \leq 3$ km. Both are run on cluster centroids and, for the `spatstat` leg, also on the individual dead trees (intensity fitted on per-cell tree counts; the point pattern uniformly subsampled to ≤20,000 per block with the null intensity rescaled to the observed count), so the result does not depend on the analysis unit.

4 Results

4.1 Cluster Characterisation

The 14,582 filtered clusters span a broad range of sizes and morphologies (Figure 2). The median cluster contains 8 trees (mean = 13.75, SD = 44.03, max = 2,965), reflecting a highly right-skewed size distribution dominated by small clusters but with a heavy tail of large events. Median area is 388 m^2 (mean = 1,061 m^2, SD = 5,003 m^2). Shapes are moderately compact (mean compactness = 0.595, SD = 0.143, median = 0.609) and moderately elongated (mean AR = 2.548, SD = 1.357, median = 2.211). The Donnelly-corrected Clark–Evans index (mean = 1.766, SD = 0.393, median = 1.753) is greater than unity, nominally indicating more-regular-than-Poisson spacing. We read this cautiously: at a median of eight points per cluster the index is high-variance, and DBSCAN's minimum-density requirement (min_samples within eps) imposes a spacing floor that mechanically inflates apparent regularity, so the value should not be over-interpreted as an ecological property of tree placement. Mean within-cluster NND is 7.79 m (SD = 2.43 m), consistent with individual-tree-scale spacing. Fractal dimension averages 1.650 (SD = 0.077).

Correlation analysis (Supplementary Fig. S1) shows strong positive covariation among size metrics (n, area) and expected trade-offs between compactness and aspect ratio. Supplementary Fig. S2 presents key pairwise scatter plots.

The spatial distribution of all 14,582 cluster centroids

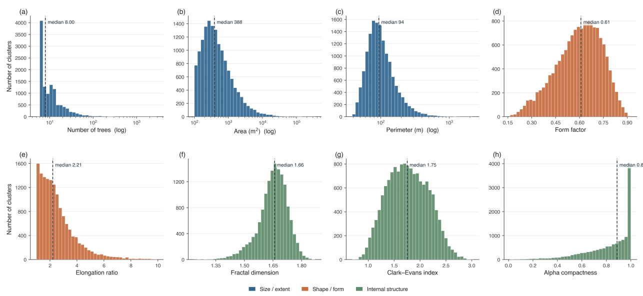

Figure 2: Distributions of key cluster metrics across all 14,582 clusters, coloured by metric family (blue: size/extent; orange: shape/form; green: internal structure). Panels (a)–(c) span two to three orders of magnitude and use a logarithmic x-axis with log-spaced bins; on this scale area and perimeter are approximately log-normal. The remaining panels use linear axes. Dashed lines mark per-metric medians.

across the sampled tiles is the sampling footprint shown earlier in Figure 1.

4.2 Morphological Gradient and Candidate Partition

K-sweep and selection. Among $k = 2, \ldots, 7$, $k = 2$ gave the best combined validation performance (silhouette $= 0.401$, CH $= 8767$, DB $= 1.073$).

Significance against a covariance-preserving null. The observed silhouette (0.401) exceeded the mean (0.321) and 95th percentile (0.324) of a 999-replicate null drawn from a single multivariate Gaussian matched to the observed feature covariance, yielding $p = 0.001$ (Figure 3). Because this null preserves all inter-feature correlation, the result indicates that the data are more structured than one elongated correlated cloud, a substantially stronger statement than rejecting the column-shuffled null, which any correlated dataset trivially beats.

Continuum, not dichotomy. Gaussian-mixture BIC decreases monotonically beyond $k = 2$ (BIC: 1.163×10^5, 1.100×10^5, 1.095×10^5, 1.085×10^5, 1.074×10^5, 1.073×10^5 for $k = 1, \ldots, 6$), so no two-component model is privileged; and the first principal component (60% of variance) is unimodal (Sarle bimodality coefficient 0.486, below the 0.555 uniform threshold). Only 2.3% of clusters lie within 5% of the $k = 2$ decision boundary, but this assignment cleanliness is expected when partitioning an elongated unimodal cloud and does not imply two natural modes. We therefore interpret the $k = 2$ partition as two endpoints of a continuous compactness–elongation gradient.

The gradient is within-band, not a latitudinal mixture. Because the sample spans 60–68°N, across which disturbance regimes differ (beetle is host- and temperature-limited toward the south; snow-load and wind rise in the centre; spruce thins toward Lapland), a pooled gradient could in principle be an artefact of mixing latitudinally-distinct regimes. It is not. Computed separately within each of the 7 sampling blocks (Table 1), the defining gradient axis is essentially identical everywhere: the within-block compactness–aspect-ratio correlation is $r \in [-0.869, -0.855]$ across all blocks, from the southern coast (60.6°N) to Lapland (67.6°N). The compactness–elongation gradient is thus a within-block morphological phenomenon reproduced at every site, not an artefact of pooling latitudinally-distinct disturbance regimes. We caution that this near-constant r is consistent with the trade-off being a geometric near-tautology (both axes are hull-derived) rather than an ecological law; what it licenses is that the *form* of the relationship is region-invariant, not that the *position* of clusters along it is representatively sampled.

Table 1: Per-block coverage. The compactness–aspect-ratio gradient axis is invariant across all seven sampling blocks; Type-0 share and block size vary. Blocks are ordered by latitude.

Block (approx. location)	n clusters	Type 0 %	r(AR, comp.)
SW coast (60.6°N)	1,243	23.6	−0.860
South (60.7°N)	4,275	22.7	−0.866
Southeast (60.7°N)	2,116	20.0	−0.864
East (61.4°N)	370	24.6	−0.855
SE inland (62.2°N)	984	28.4	−0.869
Central-north (65.1°N)	4,217	22.8	−0.867
Lapland (67.6°N)	1,377	29.3	−0.868

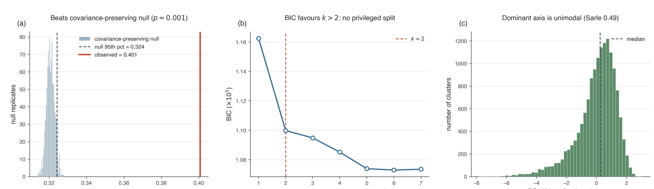

Figure 3: Significance and discreteness of the $k = 2$ partition. **(a)** The observed silhouette (red line, 0.401) exceeds a 999-replicate *covariance-preserving* null (single multivariate Gaussian matched to the feature covariance; mean 0.321, 95th pct 0.324), $p = 0.001$, a far stricter test than column-shuffling. **(b)** Gaussian-mixture BIC decreases monotonically past $k = 2$, so no two-component model is privileged. **(c)** The first principal component (60% of variance) is unimodal (Sarle coefficient 0.486). Together: real structure, but a continuous gradient rather than two discrete classes.

Endpoint descriptions. Table 2 summarises per-endpoint metric means and standard deviations. The *elongated/dispersed* endpoint (*Type 0*; 3,422 clusters, 23.5%) is characterised by high aspect ratio (4.291±1.490), low compactness (0.410±0.093), and elevated Clark–Evans index (2.044±0.360). The *compact/aggregated* endpoint (*Type 1*; 11,160 clus-

Table 2: Per-type means ± SD for all morphological features. Clustering was performed on fractal_dim, clark_evans, and aspect_ratio only; the remaining features are descriptive.

Feature	Type 0 (elongated/dispersed) ($n = 3,422$)	Type 1 (compact/aggregated) ($n = 11,160$)
Fractal dimension	1.555 ± 0.069	1.679 ± 0.051
Clark–Evans index	2.044 ± 0.360	1.681 ± 0.363
Aspect ratio	4.291 ± 1.490	2.014 ± 0.714
Compactness	0.410 ± 0.093	0.651 ± 0.101
Alpha compactness	0.713 ± 0.250	0.855 ± 0.161
Core-to-edge ratio	0.662 ± 0.104	0.799 ± 0.089
Mean NND (m)	8.555 ± 2.489	7.561 ± 2.360
Mean area (m^2)	382	1,269
Mean n (trees)	7.7	15.6

ters, 76.5%) shows rounder, denser shapes (compactness = 0.651±0.101, AR = 2.014±0.714) with higher fractal dimension (1.679±0.051) and larger mean area (1,269 m^2 vs. 382 m^2). We use *Type 0 / Type 1* purely as labels for the two ends of the gradient and attach no disturbance-agent meaning to them here.

The spatial distribution of the two gradient endpoints within a southern display window (one tile, 10.9% of clusters, shown for legibility) is given in Supplementary Fig. S3; the inset locates this window within the national sample.

Size-frequency distributions (E1). For both endpoints a lognormal cluster-size distribution is preferred over a power law (likelihood-ratio test $p < 0.001$; Clauset et al. 23). We therefore do not interpret the nominal power-law exponents as process signatures, once the power law is rejected, its exponent is not a meaningful mechanistic parameter. Descriptively, Type 1 has the heavier upper tail (Supplementary Fig. S4), but this follows directly from its larger mean cluster area (1,269 vs. 382 m^2) and does not by itself imply a contagious generating process.

Position in shape space (E5). Figure 4 plots each cluster in (compactness, aspect_ratio) space against a set of *illustrative* morphological regions loosely informed by qualitative descriptions of wind, beetle, and fire patches in the literature [9; 12; 6]. We stress that these regions are author-defined bounding boxes, not published (compactness, aspect_ratio) envelopes, the cited studies do not report cluster-level shape coordinates by agent, and that any apparent separation is partly circular: aspect ratio is a clustering input and compactness is geometrically coupled to it. The descriptive overlaps (Table 3) therefore restate the gradient definition rather than provide independent ecological validation, and we report them only to locate the two endpoints in a shape space comparable to qualitative literature accounts. Establishing genuine agent correspondence requires overlay with an independent disturbance-agent map (see Discussion).

Table 3: Position of each endpoint relative to illustrative, author-defined shape regions (E5). These regions are not published reference envelopes and the overlaps are partly circular (see text); they are descriptive, not validating.

Illustrative region	Type 0	Type 1
"Elongated" (wind-like)	57.0%	1.2%
"Intermediate" (beetle-like)	75.0%	66.4%
"Compact" (fire-like)	32.3%	94.7%

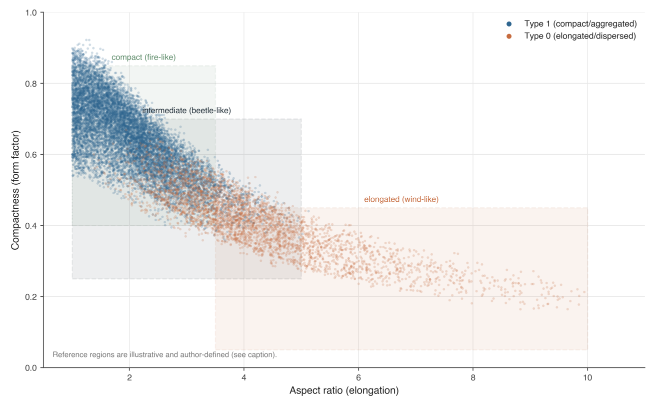

Figure 4: Shape-space scatter (compactness vs. aspect ratio) per endpoint, with *illustrative, author-defined* shape regions overlaid. The separation is partly circular because aspect ratio defines the gradient and compactness is coupled to it; the regions are not published per-agent envelopes.

4.3 Orientation Analysis

Rayleigh tests on doubled orientation angles show no meaningful preferred direction in either type (Figure 5). For Type 0: $r = 0.009$, mean direction = 156.4°, $p = 0.747$. For Type 1: $r = 0.040$, mean direction = 147.0°, $p = 2.0 \times 10^{-8}$. Although the Type 1 result is statistically significant given the large sample ($n = 11{,}160$), the mean resultant length $r \approx 0.04$ is ecologically trivial: orientations are effectively uniformly distributed in both types. The null orientation finding for Type 0 is discussed in Section 5.

4.4 Spatial Autocorrelation

Figure 6 presents the multi-scale Moran's I heatmap (E6) for four metrics at five distance thresholds. The autocorrelation that is present is weak in magnitude ($I \leq 0.11$ throughout). We are deliberately conservative about its significance, its robustness, and its spatial support, for three reasons. First, under the analytical p-value only Clark–Evans clears the Bonferroni threshold, and only at the 500 m–5 km scales; the fractal-dimension, aspect-ratio, and compactness autocorrelations are not Bonferroni-significant. Second, the detection-robustness gate shows that the *sign* of I survives a 15% false-

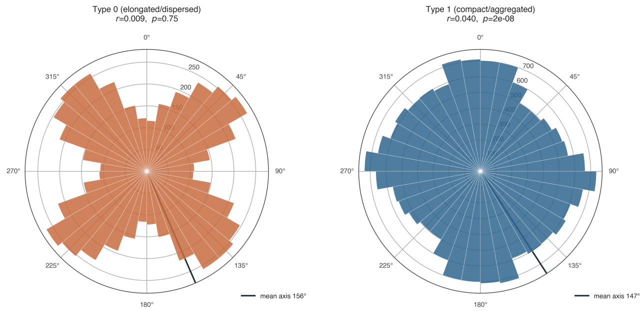

Figure 5: Polar rose plots of cluster orientation per type (axial data, 10° bins). Neither type shows a meaningful preferred direction ($r < 0.05$ in both cases).

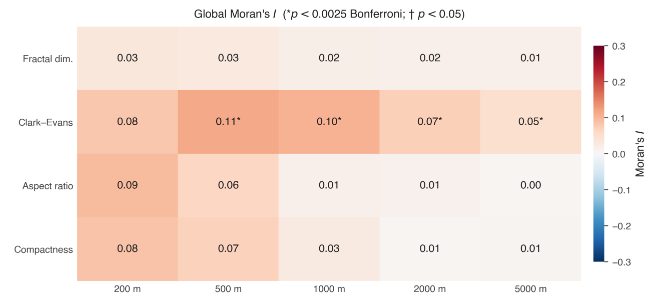

Figure 6: Moran's I heatmap (E6): four metrics (rows) $\times$ five distance thresholds (columns), analytical p. Cells marked $*$ clear the Bonferroni threshold $\alpha^* = 0.0025$, † are nominally significant ($p < 0.05$). Autocorrelation is weak throughout ($I \leq 0.11$); only Clark–Evans is Bonferroni-significant (1–5 km), and a 15% false-negative detection perturbation leaves the sign of I stable for all metrics except fractal dimension. The 200 m column rests on $\sim$20% of points (80% islands); km-scale columns are well supported. Weights connect within-band neighbours only.

negative detection perturbation for Clark–Evans, aspect ratio, and compactness ($\geq 96\%$ of realizations) but *not* for fractal dimension (sign preserved in only 60–72%), whose near-zero autocorrelation is indistinguishable from noise once detection error is propagated. Third, and importantly given the banded sample, the short-distance thresholds rest on very thin support: at 200 m, 80% of the subsample clusters are islands (no neighbour within threshold, mean degree 0.3), falling to 52% at 500 m and 0.8% at 5 km (mean degree 29). The nominal "strongest at 200 m" cells are therefore computed on $\sim$20% of points, whereas the km-scale cells are well supported; this is the opposite of the usual concern that the largest threshold is the sparsest. The defensible statement is therefore narrow: within-cluster dispersion (Clark–Evans) shows weak but robust positive spatial autocorrelation at the well-supported 1–5 km scale, while the shape metrics show at most marginal, detection-sensitive structure. Because the sample is in discrete latitudinal bands separated by large gaps, all distance-band weights connect only within-band neighbours: the reported autocorrelation is a *within-band* quantity, and between-band (latitudinal) structure is unsampled rather than absent. LISA maps (Supplementary Fig. S5) illustrate the spatial structure of fractal-dimension autocorrelation within the southern display window; labels are computed on the full survey dataset to preserve correct neighbourhood weights, and given the weak global signal for this metric they are best read qualitatively.

4.5 Landscape Connectivity

At the 200 m threshold, the proximity graph over all clusters contains 13,351 edges; the largest connected component spans only 175 of 14,582 clusters and roughly a third of clusters are isolated singletons, so the graph is sparse. Supplementary Fig. S6 shows connectivity reach and bridge clusters within the southern display window (statistics from the full graph). Pooled across all clusters, Type 1 appears more connected than Type 0 (median reach = 3 vs. 2; mean reach 11.03 vs. 8.98; Mann–Whitney $p = 1.7 \times 10^{-10}$), but this is a pseudoreplication artefact: the rank-biserial effect size is only $r = 0.07$, and the test treats 14,582 block-nested clusters as independent. Tested honestly at the level of the 7 sampling blocks, the difference does not survive: only 2 of 7 blocks show Type 1 > Type 0 median reach (sign test $p = 0.45$). The pooled "significance" was entirely a large-n effect, and what difference exists is a direct consequence of Type 1's larger area and denser packing (larger, denser clusters are mechanically more likely to fall within 200 m of a neighbour). We therefore treat reach as a density correlate, not an independent line of evidence. Because the graph is untimed and unweighted, we do not interpret high-betweenness clusters as outbreak corridors or sentinels; they are geometric articulation points whose ecological relevance is untested. We also note that, because the sample is in discrete blocks separated by far more than 200 m, no connected component can bridge blocks: reach is a strictly within-block measure of local packing density and carries no between-region connectivity meaning by construction. A meaningful connectivity analysis would require a graph whose length scale is matched to a dispersal process, for a bark-beetle hypothesis, a host-conditioned, dispersal-weighted graph at the $\sim$250 m–1 km scale of *Ips typographus* flight; for wind, storm-front co-occurrence at the tens-of-kilometre scale of a storm footprint, and these are mutually incompatible graphs, which is itself why a single undirected 200 m graph cannot adjudicate either process.

4.6 Feature-Space Coherence Check

The held-out Random Forest classifier achieved AUC $= 0.976 \pm 0.002$ (range 0.973–0.978 across 5 spatial folds; Supplementary Fig. S7); re-running the cross-validation with the 7 sampling blocks as the grouping variable (leave-one-block-out), which matches the true spatial-dependence structure, gives an essentially identical AUC = 0.975 ± 0.003. We caution against reading this as independent validation. The "held-out" features are geometrically coupled to the K-means inputs: compactness, for example, correlates with aspect ratio at $r = -0.865$ and alone attains a univariate AUC of about 0.96, so a high AUC is largely a restatement of the K-means boundary through correlated geometry rather than corroboration from new information. We report it only as an internal coherence check; on an imbalanced 76.5/23.5 partition, ROC-AUC should in future be complemented by precision–recall AUC and per-class recall.

SHAP mean absolute values ranked features as follows: compactness (0.226) > core_to_edge_ratio (0.068) > nnd_mean (0.037) > orientation (0.034) > alpha_compactness (0.019) > nnd_std (0.010) > reach_200m (0.005). Compactness dominates because it is geometrically correlated with aspect ratio (a K-means input). Crucially, reach_200m, the only feature in this set that is independent of cluster shape, ranks last, confirming that high AUC is primarily driven by correlated shape dimensions rather than landscape connectivity.

This same asymmetry is what licenses reporting the gradient while withholding the discrete typology. The detector confound (G4) predicts the *discrete* type at AUC = 0.727, yet a cluster's position along the *continuous* gradient (first principal component of the three features) correlates with detection confidence at only $\rho = 0.07$. The discrete labels sit close to the detector's decision surface; the continuous morphospace coordinate does not. The gradient is therefore the level at which the signal is least entangled with the instrument, and the typology is the level at which it is most entangled, hence the former is reported and the latter withheld.

Because the sample spans 60–68°N, we further checked whether this detector confound is merely a latitudinal gradient (canopy composition, solar geometry, and snow/phenology all vary with latitude and could structure detector recall). It is not: cluster location alone (easting, northing) predicts type at AUC = 0.519, essentially chance, consistent with the near-constant Type 0 prevalence across latitude bands (22.9%, 22.8%, 29.3% from south to north), and the detector confound is unchanged when location is partialled out (AUC $0.727 \rightarrow 0.720$). The confound is thus driven by detection covariates (confidence, density), not by geography, and the morphological gradient is not a latitudinal climate proxy.

4.7 Stability Gates

The four gates jointly indicate that the $k = 2$ split is not a discrete typology but a convenience cut through a continuous gradient, and that one of its endpoint labels is unsupported.

(G1) The partition is not scale-stable. Re-clustering from raw across eps $\in \{10, \ldots, 40\}$ m, the Type 0 fraction slides monotonically (28.6% at 10 m, 25.8%, 23.5% at 20 m, 24.0%, 23.4%, 22.9% at 40 m) and the Type 0 centroid drifts continuously rather than holding position (aspect ratio $2.97 \rightarrow 4.29 \rightarrow 4.79$; compactness $0.544 \rightarrow 0.410 \rightarrow 0.372$ across eps 10/20/40 m). A genuine class would be eps-invariant; this rescaling with the clustering radius is the signature of detection chaining, not a fixed disturbance class. BIC favours more than two components at every eps.

(G2) The headline prevalence is an artefact of the AR cap. Re-clustering from raw at eps = 20 m with no AR cap (14,644 clusters) and sweeping the cap shows the Type 0 fraction is conditional on where the cap falls: 33.1% (AR ≤ 5), 27.0% (≤ 7), 23.5% (≤ 10, the published value), 21.8% (≤ 15), down to 21.3% (uncapped). The within-Type 0 AR distribution is pressed against the cap (maximum 9.94 at AR ≤ 10) and, once released, extends as a single smooth tail to AR ≈ 30 with no second mode. Membership ARI against the AR ≤ 10 partition decays monotonically as the tail is restored ($1.00 \rightarrow 0.92 \rightarrow 0.89$). Aspect ratio is therefore a continuum, and the AR ≤ 10 cap manufactured the clean "23.5%" figure (Figure 8 summarises G1 and G2).

(G3) No directional signal beyond hull shape. Within matched hull-AR bins, Type 0 and Type 1 have statistically indistinguishable point-scatter linearity (Figure 7; e.g. AR $\in (2, 2.5]$: 0.829 vs. 0.830, $p = 0.86$; AR $\in (3, 4]$: 0.916 vs. 0.915, $p = 0.73$; Type 1 marginally higher in the $(2.5, 3]$ bin). The overall impression that Type 0 is "more linear" (Figure 7a) is entirely a hull-AR effect; once hull elongation is controlled (Figure 7b) the elongated endpoint carries no extra line-like (wind-swath) structure. Together with the orientation null (Section 4.3), this empirically removes the basis for a "wind-thrown" interpretation.

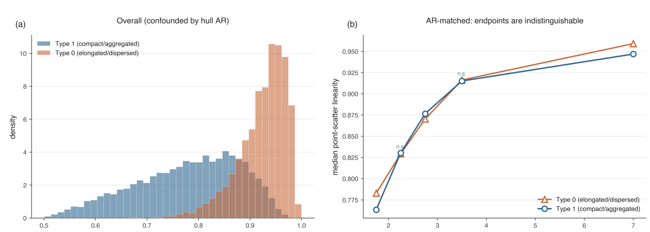

Figure 7: Second-order anisotropy gate (G3). **(a)** Overall, the elongated endpoint appears more linear, but this is confounded by hull aspect ratio. **(b)** Matched on hull aspect ratio, the two endpoints' median point-scatter linearity is indistinguishable (n.s. where shown), so the elongated endpoint carries no directional point-process signal beyond hull shape.

(G4) A measurable detector confound. Per-cluster detection-confidence covariates predict type at

AUC = 0.727 (5-fold), and dropping 10–20% of detections per cluster flips the type of 5.9–7.5% of clusters (ARI 0.76–0.70). The typology is thus partly entangled with detector behaviour; cleanly separating the two requires stratified detection precision/recall against ground truth, which the present data lack.

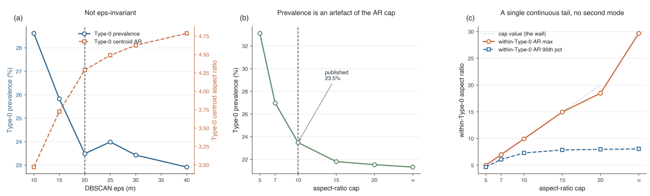

Figure 8: Stability gates. The Type 0 fraction is not invariant to DBSCAN eps (G1) and is conditional on the aspect-ratio cap (G2); the elongated endpoint's AR distribution forms a single continuous tail once the cap is released, with no second mode.

4.8 Host-Conditioned Aggregation of Mortality

This is the primary analysis. Fitting the pixel-Poisson intensity model on the forest grid, mortality intensity increases with *both* spruce and total growing-stock volume (standardised log-linear coefficients +0.23 and +0.21 respectively; stand-edge proximity negligible, −0.02): canopy mortality strongly tracks forest host availability, as expected.

Beyond this host-tracking, mortality is *still* aggregated more than the host predicts at short range. Against a spruce-only host null, the inhomogeneous L-function (a standard measure of how clustered points are, here calibrated against the host surface rather than against an even scatter) lies far above the host-conditioned envelope (the band of values expected if mortality merely followed host) across 100 m–1 km (Figure 10a). Crucially, the signal survives the richer null: with intensity conditioned on spruce, total volume, and stand edge, the observed $L_{\text{inhom}}(r)-r$ remains above the simulation envelope across the same range (Figure 10b), although the excess is much smaller: adding total volume absorbs most of the apparent "foci beyond spruce," confirming that a large part of the spruce-only residual is broader forest structure that the spruce layer alone does not resolve. What remains is a *modest but genuine residual aggregation at the sub-kilometre scale*: a local-contagion signature on top of host-tracking, consistent with mortality concentrating into foci rather than spreading diffusely across available host. Detection error does not explain it: per-cluster detector confidence predicts cluster "type" at AUC = 0.727 but the continuous structure couples to confidence only weakly ($\rho = 0.07$).

Independent confirmation (spatstat). We reproduced the analysis in a second, independent implementation: a per-block inhomogeneous L-function in `spatstat`, with the intensity fitted globally (a pooled Poisson model across all forest cells, capturing the regional host gradient) and applied within each block, and significance from host-conditioned simulation envelopes (the estimator was verified on simulated inhomogeneous-Poisson patterns). The result is consistent in *all seven* blocks: the observed $L_{\text{inhom}}(r)-r$ lies above the host-conditioned envelope across 250 m–1 km (the centroid curves of Figure 9). The two implementations agree on the qualitative finding (residual aggregation beyond host, present in every block) and on the robustness of spruce-driven host-tracking; they differ in the *magnitude* of the residual (the per-block cumulative L accumulates more multi-scale clustering than the tile-pooled pair-correlation), so we report the existence and scale of the residual, not a single magnitude.

Beyond the centroid, beyond 1 km. Two pure-computation refinements confirm that the residual is neither an artefact of the clustering unit nor confined to short range. *(i) Tree-level.* Re-running the `spatstat` analysis on the individual dead trees (intensity fitted on per-cell tree counts; the point pattern uniformly subsampled to ≤20,000 per block with the null intensity rescaled to the observed count) leaves the conclusion unchanged: $L_{\text{inhom}}(r)-r$ lies above the host-conditioned envelope in all seven blocks across 250 m–2 km (Figure 9), with the tree-count intensity model tracking host even more strongly than the cluster model (standardised spruce +0.30, total volume +0.25). *(ii) Beyond 1 km.* The 25–54 km block windows support a cumulative, translation-corrected estimator well past the 1 km ceiling of the 6 km tile-pooled ring estimator; extending the centroid analysis to $r = 3$ km, the exceedance persists in every block out to 3 km, either plateauing (blocks 0–2, 4) or continuing to rise (the multi-scale clustering of blocks 5–6). The residual aggregation is therefore established at both the cluster and the individual-tree level, across 250 m to at least 3 km, on the 7-block sample.

5 Discussion

What controls where dead trees concentrate? The central result is the host-conditioned one (Section 4.8): boreal canopy mortality strongly tracks forest host availability (intensity rises with both spruce and total growing-stock volume) yet retains a modest, genuine residual aggregation, robust to the analysis unit (cluster or individual tree) and persisting from the sub-kilometre scale out to at least 3 km, that survives a richer host-conditioned null. Ecologically this is the signature of a process that is *first* constrained by where susceptible host stands are, and *then* concentrates into local foci within them rather than spreading diffusely, consistent with, though not proof of, short-range contagion (bark-beetle tree-to-tree spread, or locally corre-

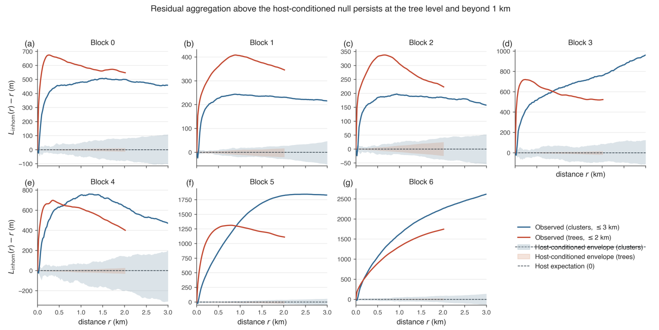

Figure 9: The residual aggregation is robust to the analysis unit and extends beyond 1 km. Per-block $L_{\text{inhom}}(r) - r$ for cluster centroids (blue, to 3 km) and individual dead trees (red, to 2 km), each against its own host-conditioned simulation envelope (shaded bands); the dashed line at 0 is the host-conditioned expectation. In all seven blocks both observed curves lie above their envelopes across the whole range, so the local-contagion signal is not an artefact of using cluster centroids and is not confined to the sub-kilometre scale.

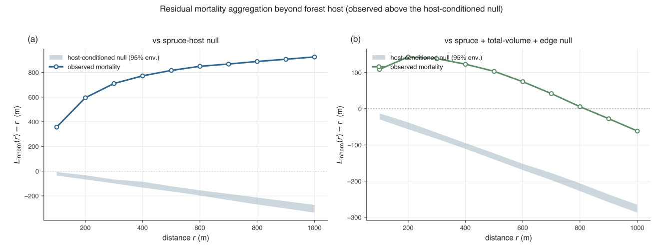

Figure 10: Residual mortality aggregation beyond forest host. Inhomogeneous $L_{\text{inhom}}(r) - r$ (observed, line) vs the host-conditioned 95% simulation envelope (band). **(a)** spruce-only host null; **(b)** richer null (spruce + total volume + stand edge). Observed lies above the envelope across 100 m–1 km in both, so the residual sub-km aggregation survives the richer host model (its magnitude shrinks as total volume absorbs broader host structure). This tile-pooled ring estimator is valid to $r \lesssim 1$ km for the 6 km tile windows; the larger-window `spatstat` estimator (Figure 9) extends the range to 3 km.

lated wind/snow damage). The honest weight of the claim is set by what conditioning removed: adding total volume to the null absorbed most of the apparent excess, so the defensible statement is a *bounded* residual, not the order-of-magnitude clustering a naive (homogeneous-null) reading suggested. This is also why we condition on a pre-mortality host and report structure rather than agents: the same morphology arises from different agents (below), so the population-level second-order signal, not patch shape, is what the data can support.

Why don't patch shapes sort into distinct disturbance types? It is tempting to map the two endpoints onto dominant Finnish disturbance agents, the elongated, internally dispersed Type 0 onto wind-felled strips [6; 7], and the compact, denser Type 1 onto radially spreading beetle kill [4; 12; 13]. We deliberately resist this mapping, because the evidence that would license it is either absent or circular. The shape-space "reference zones" are author-defined boxes in a space whose axes (aspect ratio, and the compactness coupled to it) *define* the gradient, so their overlap with the endpoints is a restatement of the clustering, not external corroboration. Several confounders independently block attribution: the detections themselves carry no species label, so a beetle-killed spruce cannot be distinguished from a drought- or pathogen-killed pine, nor a compact patch from fire, snow-load, or small salvage/clearcut polygons; the `alku_aika` field records flight-strip acquisition time, not mortality date, severing any link to specific storms or outbreak years; and the single 2023 snapshot cannot separate a recent event from a years-old palimpsest of snags. A compact morphology, in particular, is the default shape of almost any small mortality patch and is therefore the least diagnostic possible signal, making a 76.5% "compact" prevalence uninformative about agent identity.

External corroboration: morphology does not track species composition. The sharpest test of the circularity is to bring in a variable from outside the shape space and the detector altogether. We attached the pre-mortality (2021) MS-NFI species composition (spruce, pine and birch growing-stock volume) to every cluster at its location and asked whether the morphological gradient covaries with the species mix of the stand that died (Section 3.10, G5). It does not. Within each block the Spearman correlation between aspect ratio and species fraction is negligible ($|\rho| \leq 0.05$ for spruce, pine and birch; Figure 11a), and the two morphological endpoints sit in near-identical species mixes (median elongated-minus-compact difference ≤ 2 percentage points for every species, with no consistent sign across blocks; Figure 11b). The only relationships whose sign is reproducible across all seven blocks, a faint tendency for more elongated patches to fall in slightly more broadleaf-rich, less coniferous stands (birch $\rho = +0.05$, conifer $\rho = -0.04$, sign-test $p = 0.016$), are an order of magnitude too small to carry ecological, let alone agent-level, meaning. An external, non-shape variable thus confirms what the internal diagnostics imply: patch morphology is not a proxy for forest composition, so it cannot be read as a proxy for the agent that a given composition would favour. This is the conservative outcome, and it strengthens the decision to withhold attribution.

Do our own analyses point to a disturbance agent, or to geometry? The most telling evidence is internal and convergent. (i) *Orientation.* The elongated endpoint shows no preferred orientation ($r = 0.009$, $p = 0.747$), yet directional alignment with storm tracks is the single most diagnostic fingerprint of windthrow. An elongated class with no net direction is precisely what one

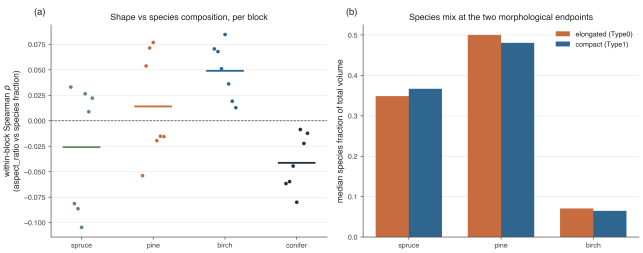

Figure 11: External validation against pre-mortality MS-NFI species composition (outside the shape space and the detector). **(a)** Within-block Spearman ρ between cluster aspect ratio and the spruce/pine/birch/conifer fraction of stand volume; points are the seven blocks, bars the across-block means. All $|\rho| \leq 0.10$ and means near zero: the morphological gradient does not track species mix (the reproducible-sign but negligible birch+/conifer− tendency aside). **(b)** Median species fraction at the compact and elongated endpoints is near-identical. Shape is not a proxy for forest composition.

would *not* expect from wind, and it is the main reason we decline the "wind-thrown" label. (ii) *Scale dependence.* The prevalence of the elongated endpoint moves with the DBSCAN radius (22.9% at 40 m to 28.6% at 10 m); a genuine ecological class should be eps-stable, whereas a chaining artefact rescales with eps. (iii) *Connectivity.* The only shape-orthogonal feature, 200 m reach, carries essentially no discriminative signal (SHAP 0.005, last of seven). Read together, non-directional elongation, eps-dependent prevalence, and no independent connectivity signal, the most parsimonious account is that the typology summarises a continuum of *detection-and-clustering geometry* (how dead- tree detections chain together at a 20 m radius, with mean within-cluster spacing of only 7.79 m), rather than two distinct disturbance processes. The earlier ecological reading of the size-frequency tails is correspondingly withdrawn: with the power law rejected in favour of a lognormal for both endpoints, the heavier Type 1 tail is explained by its larger mean area, not by contagious spread.

What does the held-out classifier's high accuracy actually mean? The AUC = 0.976 shows that the two endpoints occupy largely non-overlapping regions of the held-out feature space, but the SHAP decomposition shows why this is not independent validation: performance is driven almost entirely by compactness and core-to-edge ratio, both geometrically correlated with aspect ratio (a K-means input). The weakest predictor, reach_200m (SHAP = 0.005), is the only feature capturing landscape connectivity independently of cluster shape; its near-zero importance means the high AUC restates the K-means boundary through correlated geometry rather than confirming it from new information. The honest reading is the negative one: on the single axis genuinely independent of shape, the two endpoints are not distinguishable.

Limitations. Beyond the attribution caveats above, several additional limitations merit note. The upstream deep-learning detection model is reported here without validation metrics (precision, recall), and gate G4 shows its confidence score alone predicts type at AUC = 0.727, so detection error, likely structured by forest type, canopy density, and flight-strip geometry, is partly entangled with the morphology we measure. Fully separating the two requires stratified detection precision/recall against ground truth, which the present data lack; this is the single most important outstanding check, and the Clark–Evans index, computed on a median of eight points per cluster, is especially sensitive to it. The dataset is a national-scale but spatially discontinuous sample (312 aerial tiles across Finland, in discrete latitudinal bands), so reported prevalences are sample statistics rather than an exhaustive national census. Gate G1 confirms that the eps = 20 m clustering scale partly determines the elongation it then measures. Finally, because standing dead trees persist for years in boreal stands, a single-year snapshot cannot date mortality: what we treat as one "cluster" may aggregate trees killed by different agents in different years that happen to be co-located, so the unit of analysis is itself only loosely defined, a fundamental limit on any morphology-to-process inference from one acquisition. Finally, the 312 tiles form a purposive two-stage cluster sample (7 super-site blocks), not a probability sample of the boreal zone, and the design is latitudinally imbalanced: 5 of the 7 blocks are southern (60–62°N), while the central (65°N) and Lapland (67.6°N) regimes are each represented by a *single, unreplicated* block. Any north–south contrast is therefore confounded with block identity: there is no way to separate a latitude effect from one block's idiosyncrasy with $n = 1$ block per northern regime. Findings generalise to the within-landscape morphology of detected mortality at the southern super-sites (where blocks are replicated); extension to central and northern Finland is provisional, and the pooled gradient is a cross-regime description rather than a within-region or national estimate. We did, however, verify that the gradient *form* is reproduced within every block (Table 1), so the principal result does not depend on the relative weighting of regions; what the design cannot support is any claim about the national frequency or distribution of mortality morphologies.

Future work. Several steps would extend the host-conditioned result, in increasing order of data requirement. The two pure-computation refinements that the analysis itself called for, a tree-level (rather than centroid) treatment of the inhomogeneous point process and a variance-stabilised, large-window estimator reaching reliably past 1 km, are carried out here (Figure 9); both confirm the residual aggregation. The remaining steps need new layers. First, with publicly available layers: overlaying a species layer from Finland's multi-source national forest inventory and a regional disturbance-agent map (e.g. Metsäkeskus damage records) would enable a marked point-process test, moving from *characterising* spatial structure to *attributing* it; we note that the pan-European attribution products [11] group wind and bark beetle into a single class and so cannot, by themselves, separate the two agents, which is why a dedicated regional layer is required. A complementary detection-validation step (stratified precision/recall against field or photo-interpreted ground truth) would calibrate, rather than merely bound, the detector confound. Second, and requiring new acquisition rather than further analysis of the present data, multi-temporal imagery (prior-year aerial surveys passed through the same detector, which are not available for this study) would add the temporal dimension: it would date mortality, separate recent deaths from years-old snags, and turn the static host-conditioned pattern into a space-time process, the definitive test of whether this year's foci predict next year's mortality. The pipeline itself is computationally scalable and, once these layers are in place, could support comparative analysis across the Nordic countries.

6 Conclusion

We have analysed 698,223 individual dead-tree detections from a 2023 national-scale aerial sample (seven boreal super-sites, southern boreal zone to Lapland) as a host-conditioned point process. Conditioning on a pre-mortality forest-inventory host surface, we find that canopy mortality strongly tracks forest host availability (intensity rises with spruce and total growing-stock volume) and yet retains a modest, genuine residual aggregation at the sub-kilometre scale that survives a richer host-conditioned null, a local-contagion signature on top of host-tracking, established with inhomogeneous second-order statistics and host-conditioned simulation envelopes. We deliberately report this spatial *structure* rather than disturbance-agent labels, because a candidate morphological "typology" of the clusters does not survive scrutiny. Using DBSCAN clustering, Donnelly-corrected Clark–Evans statistics, and a suite of complementary shape metrics, the morphology of detected mortality clusters is organised by a dominant compactness–elongation gradient. This gradient is not a statistical artefact of correlated features: the $k = 2$ silhouette (0.401) exceeds a covariance-preserving null at $p = 0.001$. The *discrete* two-class reading of that gradient, however, does not survive scrutiny. Mixture-model selection and the unimodality of the dominant axis already indicate a continuum rather than two modes; four stability gates confirm it. Re-clustering from raw, the partition is not invariant to the DBSCAN radius (G1), its 23.5%/76.5% split is an artefact of the aspect-ratio filter and dissolves into a single continuous tail once that filter is released (G2), the elongated class carries no directional structure beyond hull shape (G3, consistent with the orientation null), and it is partly predictable from detector confidence alone (G4). The held-out Random Forest (AUC = 0.976) reproduces the split only through geometric redundancy among shape descriptors, the one shape-orthogonal feature, landscape connectivity, adds no signal.

The framework demonstrates that remote-sensing-derived maps of dead-tree locations contain a recoverable and reproducible morphological gradient, but that the temptation to discretise it into disturbance types must be resisted on the present evidence. The discrete partition is an artefact of clustering scale and a filter applied to its own defining axis, with no second-order content and a measurable detector confound; the gradient is what is real. Rather than a ready "first filter" for the field, the pipeline is best positioned as a hypothesis-generating characterisation whose ecological meaning must be established by overlay with independent disturbance-agent maps, species layers, and multi-temporal imagery, a programme we outline as the natural next step.

Data and Code Availability

The dead-tree detections analysed here were produced by the U-Net segmentation model of Junttila et al. [1] applied to 2023 National Land Survey of Finland aerial imagery; the derived cluster-level table (14,582 clusters with shape, point-pattern, and connectivity metrics) and the per-tree coordinates are released with the analysis code. The host and species covariates are the public multi-source national forest inventory (MS-NFI / MVMI) growing-stock volume and land-class rasters of the Natural Resources Institute Finland (Luke; CC-BY 4.0, 16 m, EPSG:3067), pre-mortality (2021) vintage: `kuusi` (spruce), `manty` (pine), `koivu` (birch), `tilavuus` (total volume) and `maaluokka` (land class), file stamp `vmi1x_1721`, retrieved from the Paituli / funet geodata archive (`rsync.nic.funet.fi/index/geodata/luke/vmi/2021/`). All analysis code (clustering, shape metrics, the four falsification gates, the host-conditioned point-process engine and its `spatstat` cross-check, and the external species validation) is available at https://github.com/aniskhan25/TreeMortSpatial. The national raster tiles are redistributed by Luke and are not duplicated in the repository; the `rsync` paths above reproduce them exactly.

Reproducibility

Analyses were run with Python 3.9.6 (numpy 2.0.2, scipy 1.13.1, pandas 2.3.3, scikit-learn 1.6.1, geopandas 1.0.1, rasterio 1.4.3, shapely 2.0.7, pyproj 3.6.1, pointpats 2.3.0, matplotlib 3.9.4) and R 4.6.1 with `spatstat` 3.6.1. All stochastic steps are seeded: the covariance-preserving null and discreteness tests use seed 42, the per-block tree subsampling for the tree-level point process uses seed 42, and the `spatstat` envelopes use 39 host-conditioned simulations per block (self-tested on inhomogeneous-Poisson patterns). DBSCAN ($\mathtt{eps} = 20\,\mathrm{m}$ for clusters, 15 km for blocks; $\mathtt{min\_samples} = 1$ for the block grouping) is deterministic. Software versions, per-figure provenance, the shape-feature correlation matrix, and univariate feature AUCs are tabulated in the Supplementary Information.

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